

Technical Report: Custom Head Coils for Neurovascular Imaging

This report details the design, electrical circuitry, and manufacturing specifications of an advanced, conformal 16-channel receive coil array optimized for high-resolution neurovascular magnetic resonance imaging. A companion .stl file (neurovascular_conformal_coil.stl) has been generated for 3D printing and CAD import.

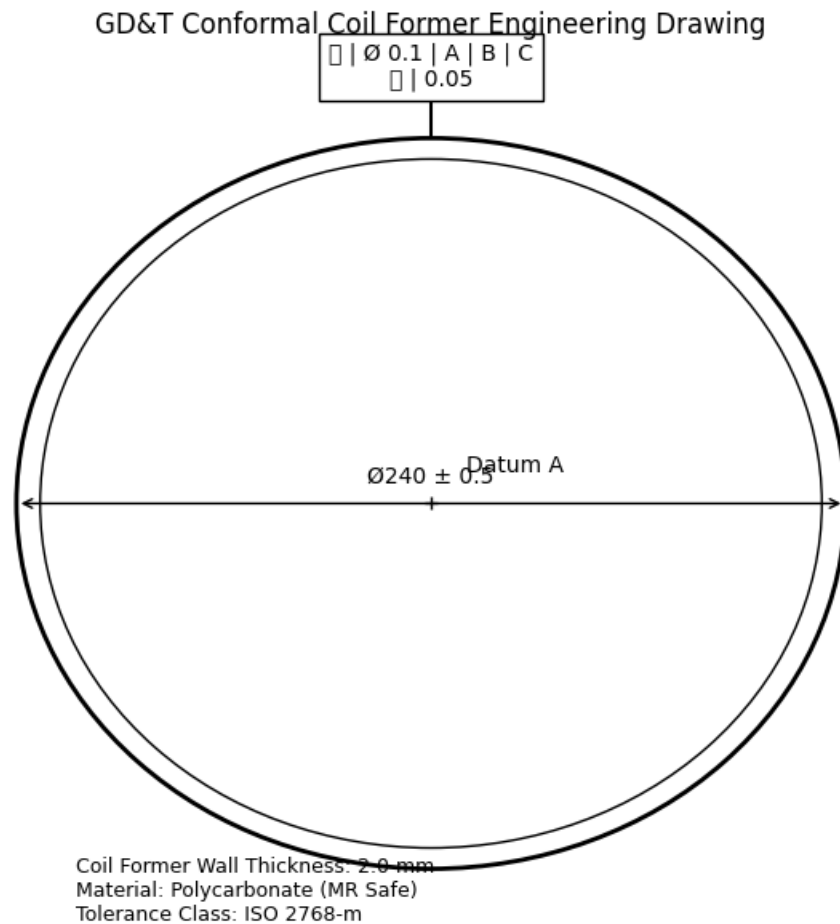


Figure 1: GD&T; Engineering Drawing for the coil former showing True Position and Cylindricity tolerances critical for maintaining conformality.

16-Channel Conformal Neurovascular Rx Coil Array

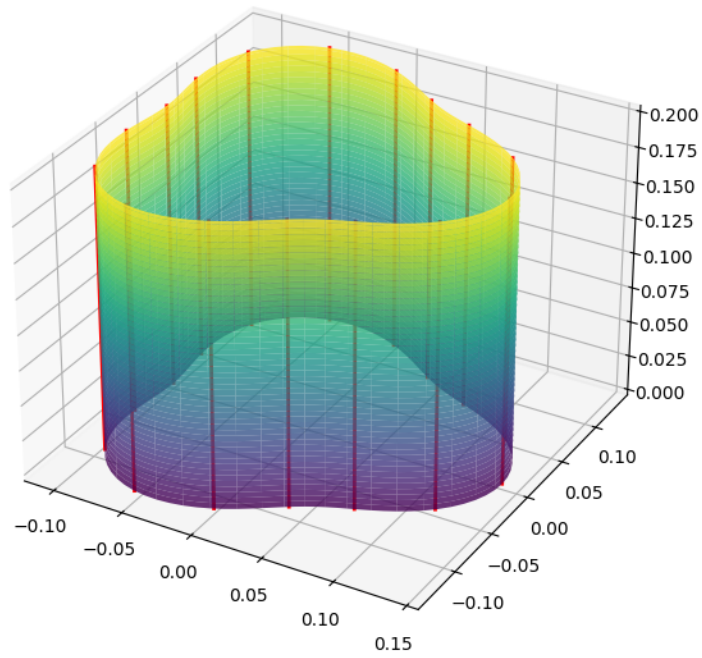


Figure 2: 3D geometry layout of the conformal coil array elements.

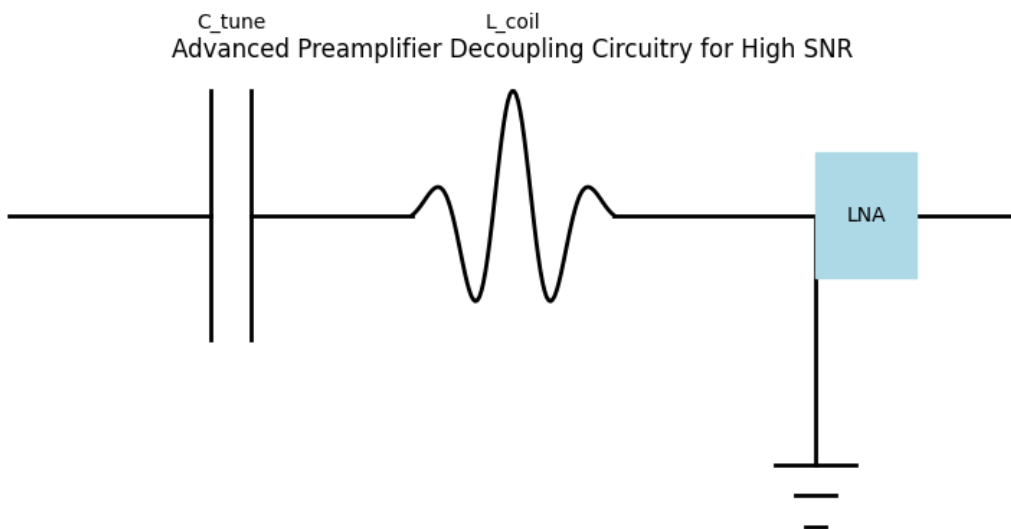


Figure 3: Electrical schematic of a single actively decoupled coil element.